Here are all 23 MSBTE Data Structures practical programs in sequential order, with filenames and plain-text code, compatible with classic Turbo C. Save each program separately using the suggested filename.

Program 1 — P01_LINEAR_NUM.C

c

```
#include<stdio.h>
#include<conio.h>

void main()
{
   int a[20],n,key,i,found=0;
   clrscr();

   printf("Enter number of elements: ");
   scanf("%d",&n);

   printf("Enter elements:\n");
   for(i=0;i<n;i++)
      scanf("%d",&a[i]);

   printf("Enter element to search: ");
   scanf("%d",&key);

   for(i=0;i<n;i++)
   {
      if(a[i]==key)
      {
         printf("Found at position %d",i+1);
         found=1;
         break;
      }
```

```
    }

    if(found==0)
        printf("Element not found");

    getch();
}
```

Program 2 — P02_LINEAR_STR.C

```
c
#include<stdio.h>
#include<conio.h>
#include<string.h>

void main()
{
    char a[10][20],key[20];
    int n,i,found=0;
    clrscr();

    printf("Enter number of strings: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%19s",a[i]);

    printf("Enter string to search: ");
    scanf("%19s",key);

    for(i=0;i<n;i++)
    {
```

```
        if(strcmp(a[i],key)==0)
        {
            printf("Found at position %d",i+1);
            found=1;
            break;
        }
    }

    if(found==0)
        printf("String not found");

    getch();
}
```

Program 3 — P03_BINARY_NUM.C

```
c
#include<stdio.h>
#include<conio.h>

void main()
{
    int a[20],n,key,low,high,mid,i;
    clrscr();

    printf("Enter number of elements: ");
    scanf("%d",&n);

    printf("Enter sorted elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);
```

```
    printf("Enter element to search: ");
    scanf("%d",&key);

    low=0;
    high=n-1;

    while(low<=high)
    {
        mid=(low+high)/2;

        if(a[mid]==key)
        {
            printf("Found at position %d",mid+1);
            getch();
            return;
        }
        else if(key>a[mid])
            low=mid+1;
        else
            high=mid-1;
    }

    printf("Element not found");
    getch();
}
```

Program 4 — P04_BINARY_STR.C

c

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
```

```
void main()
{
    char a[10][20],key[20];
    int n,low,high,mid,i,c;
    clrscr();

    printf("Enter number of strings: ");
    scanf("%d",&n);

    printf("Enter strings alphabetically:\n");
    for(i=0;i<n;i++)
        scanf("%19s",a[i]);

    printf("Enter string to search: ");
    scanf("%19s",key);

    low=0;
    high=n-1;

    while(low<=high)
    {
        mid=(low+high)/2;
        c=strcmp(a[mid],key);

        if(c==0)
        {
            printf("Found at position %d",mid+1);
            getch();
            return;
        }
        else if(c<0)
            low=mid+1;
```

```
        else
            high=mid-1;
    }

    printf("String not found");
    getch();
}
```

Program 5 — P05_BUBBLE_NUM.C

c

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int a[20],n,i,j,temp;
    clrscr();

    printf("Enter number of elements: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    for(i=0;i<n-1;i++)
    {
        for(j=0;j<n-i-1;j++)
        {
            if(a[j]>a[j+1])
            {
                temp=a[j];
```

```
            a[j]=a[j+1];
            a[j+1]=temp;
          }
       }
    }

    printf("Sorted array:\n");
    for(i=0;i<n;i++)
       printf("%d ",a[i]);

    getch();
}
```

Program 6 — P06_BUBBLE_STR.C

c

```
#include<stdio.h>
#include<conio.h>
#include<string.h>

void main()
{
    char a[10][20],temp[20];
    int n,i,j;
    clrscr();

    printf("Enter number of strings: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
       scanf("%19s",a[i]);
```

```
    for(i=0;i<n-1;i++)
    {
        for(j=0;j<n-i-1;j++)
        {
            if(strcmp(a[j],a[j+1])>0)
            {
                strcpy(temp,a[j]);
                strcpy(a[j],a[j+1]);
                strcpy(a[j+1],temp);
            }
        }
    }

    printf("Sorted strings:\n");
    for(i=0;i<n;i++)
        printf("%s\n",a[i]);

    getch();
}
```

Program 7 — P07_SELECTION_NUM.C

c

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int a[20],n,i,j,min,temp;
    clrscr();

    printf("Enter number of elements: ");
```

```
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    for(i=0;i<n-1;i++)
    {
        min=i;

        for(j=i+1;j<n;j++)
            if(a[j]<a[min])
                min=j;

        temp=a[i];
        a[i]=a[min];
        a[min]=temp;
    }

    printf("Sorted array:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    getch();
}
```

Program 8 — P08_SELECTION_STR.C

c

```
#include<stdio.h>
#include<conio.h>
#include<string.h>
```

```
void main()
{
    char a[10][20],temp[20];
    int n,i,j,min;
    clrscr();

    printf("Enter number of strings: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%19s",a[i]);

    for(i=0;i<n-1;i++)
    {
        min=i;

        for(j=i+1;j<n;j++)
            if(strcmp(a[j],a[min])<0)
                min=j;

        strcpy(temp,a[i]);
        strcpy(a[i],a[min]);
        strcpy(a[min],temp);
    }

    printf("Sorted strings:\n");
    for(i=0;i<n;i++)
        printf("%s\n",a[i]);

    getch();
}
```

Program 9 — P09_INSERTION_NUM.C

c

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int a[20],n,i,j,key;
    clrscr();

    printf("Enter number of elements: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    for(i=1;i<n;i++)
    {
        key=a[i];
        j=i-1;

        while(j>=0 && a[j]>key)
        {
            a[j+1]=a[j];
            j--;
        }

        a[j+1]=key;
    }

    printf("Sorted array:\n");
    for(i=0;i<n;i++)
```

```
        printf("%d ",a[i]);

    getch();
}
```

Program 10 — P10_SLL_BEGIN_SEARCH.C

c

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node *head=NULL;

void insert()
{
    struct node *p;
    p=(struct node*)malloc(sizeof(struct node));

    if(p==NULL)
        return;

    printf("Enter data: ");
    scanf("%d",&p->data);

    p->next=head;
```

```
    head=p;
}

void search()
{
    struct node *p=head;
    int key,pos=1;

    printf("Enter element: ");
    scanf("%d",&key);

    while(p!=NULL)
    {
        if(p->data==key)
        {
            printf("Found at position %d",pos);
            return;
        }

        p=p->next;
        pos++;
    }

    printf("Not found");
}

void display()
{
    struct node *p=head;

    while(p!=NULL)
    {
        printf("%d -> ",p->data);
```

```
        p=p->next;
    }

    printf("NULL");
}

void main()
{
    int ch;
    clrscr();

    do
    {
        printf("\n1.Insert Beginning\n2.Search");
        printf("\n3.Display\n4.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1: insert(); break;
            case 2: search(); break;
            case 3: display(); break;
        }
    }while(ch!=4);

    getch();
}
```

Program 11 — P11_SLL_END_DELETE.C

c

```
#include<stdio.h>
```

```
#include<conio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node *head=NULL;

void insertEnd()
{
    struct node *p,*t;

    p=(struct node*)malloc(sizeof(struct node));
    if(p==NULL)
        return;

    printf("Enter data: ");
    scanf("%d",&p->data);
    p->next=NULL;

    if(head==NULL)
        head=p;
    else
    {
        t=head;
        while(t->next!=NULL)
            t=t->next;
        t->next=p;
    }
}
```

```
void deleteBegin()
{
    struct node *t;

    if(head==NULL)
    {
        printf("List empty");
        return;
    }

    t=head;
    printf("Deleted: %d",t->data);
    head=head->next;
    free(t);
}

void display()
{
    struct node *t=head;

    while(t!=NULL)
    {
        printf("%d -> ",t->data);
        t=t->next;
    }

    printf("NULL");
}

void main()
{
    int ch;
```

```
    clrscr();

    do
    {
        printf("\n1.Insert End\n2.Delete Beginning");
        printf("\n3.Display\n4.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1: insertEnd(); break;
            case 2: deleteBegin(); break;
            case 3: display(); break;
        }
    }while(ch!=4);

    getch();
}
```

Program 12 — P12_SLL_INSERT_AFTER.C

c

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node *next;
};
```

```
struct node *head=NULL;

void insertEnd()
{
    struct node *p,*t;

    p=(struct node*)malloc(sizeof(struct node));
    if(p==NULL)
        return;

    printf("Enter data: ");
    scanf("%d",&p->data);
    p->next=NULL;

    if(head==NULL)
        head=p;
    else
    {
        t=head;
        while(t->next!=NULL)
            t=t->next;
        t->next=p;
    }
}

void insertAfter()
{
    struct node *p,*t=head;
    int key;

    printf("Insert after which value? ");
    scanf("%d",&key);
```

```
    while(t!=NULL && t->data!=key)
        t=t->next;

    if(t==NULL)
    {
        printf("Element not found");
        return;
    }

    p=(struct node*)malloc(sizeof(struct node));
    if(p==NULL)
        return;

    printf("Enter new data: ");
    scanf("%d",&p->data);

    p->next=t->next;
    t->next=p;
}

void display()
{
    struct node *t=head;

    while(t!=NULL)
    {
        printf("%d -> ",t->data);
        t=t->next;
    }

    printf("NULL");
}
```

```
void main()
{
   int ch;
   clrscr();

   do
   {
      printf("\n1.Insert End\n2.Insert After");
      printf("\n3.Display\n4.Exit\n");
      scanf("%d",&ch);

      switch(ch)
      {
         case 1: insertEnd(); break;
         case 2: insertAfter(); break;
         case 3: display(); break;
      }
   }while(ch!=4);

   getch();
}
```

Program 13 — P13_STACK_ARRAY.C

c

```
#include<stdio.h>
#include<conio.h>
#define MAX 5

int stack[MAX],top=-1;

void push()
```

```
{
    int x;

    if(top==MAX-1)
    {
        printf("Stack Overflow");
        return;
    }

    printf("Enter element: ");
    scanf("%d",&x);
    stack[++top]=x;
}

void pop()
{
    if(top==-1)
        printf("Stack Underflow");
    else
        printf("Popped: %d",stack[top--]);
}

void display()
{
    int i;

    for(i=top;i>=0;i--)
        printf("\n%d",stack[i]);
}

void main()
{
    int ch;
```

```
    clrscr();

    do
    {
        printf("\n1.Push\n2.Pop\n3.Display\n4.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1: push(); break;
            case 2: pop(); break;
            case 3: display(); break;
        }
    }while(ch!=4);

    getch();
}
```

Program 14 — P14_STACK_LINKED.C

c

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node *top=NULL;
```

```
void push()
{
    struct node *p;
    p=(struct node*)malloc(sizeof(struct node));

    if(p==NULL)
        return;

    printf("Enter element: ");
    scanf("%d",&p->data);

    p->next=top;
    top=p;
}

void pop()
{
    struct node *t;

    if(top==NULL)
    {
        printf("Stack Underflow");
        return;
    }

    t=top;
    printf("Popped: %d",t->data);
    top=top->next;
    free(t);
}

void display()
```

```
{
    struct node *t=top;

    while(t!=NULL)
    {
        printf("%d -> ",t->data);
        t=t->next;
    }

    printf("NULL");
}

void main()
{
    int ch;
    clrscr();

    do
    {
        printf("\n1.Push\n2.Pop\n3.Display\n4.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1: push(); break;
            case 2: pop(); break;
            case 3: display(); break;
        }
    }while(ch!=4);

    getch();
}
```

Program 15 — P15_RECURSIVE_MULTIPLY.C

c

```
#include<stdio.h>
#include<conio.h>

int multiply(int a,int b)
{
    if(b==0)
        return 0;

    if(b<0)
        return -multiply(a,-b);

    return a+multiply(a,b-1);
}

void main()
{
    int a,b;
    clrscr();

    printf("Enter two numbers: ");
    scanf("%d%d",&a,&b);

    printf("Product = %d",multiply(a,b));

    getch();
}
```

Program 16 — P16_REVERSE_RECURSION.C

c

```
#include<stdio.h>
#include<conio.h>
#include<string.h>

void reverse(char s[],int i)
{
    if(i<0)
        return;

    printf("%c",s[i]);
    reverse(s,i-1);
}

void main()
{
    char s[50];
    clrscr();

    printf("Enter string: ");
    scanf("%49s",s);

    printf("Reverse string: ");
    reverse(s,strlen(s)-1);

    getch();
}
```

Program 17 — P17_LINEAR_QUEUE_ARRAY.C

c

```
#include<stdio.h>
#include<conio.h>
#define MAX 5

int q[MAX],front=-1,rear=-1;

void insert()
{
    int x;

    if(rear==MAX-1)
    {
        printf("Queue Overflow");
        return;
    }

    printf("Enter element: ");
    scanf("%d",&x);

    if(front==-1)
        front=0;

    q[++rear]=x;
}

void delete()
{
    if(front==-1)
    {
        printf("Queue Underflow");
        return;
    }
```

```
    printf("Deleted: %d",q[front++]);

    if(front>rear)
        front=rear=-1;
}

void display()
{
    int i;

    for(i=front;i!=-1 && i<=rear;i++)
        printf("%d ",q[i]);
}

void main()
{
    int ch;
    clrscr();

    do
    {
        printf("\n1.Insert\n2.Delete\n3.Display\n4.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1: insert(); break;
            case 2: delete(); break;
            case 3: display(); break;
        }
    }while(ch!=4);

    getch();
```

```
}
```

Program 18 — P18_LINEAR_QUEUE_LINKED.C

c

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node *next;
};

struct node *front=NULL,*rear=NULL;

void insert()
{
    struct node *p;
    p=(struct node*)malloc(sizeof(struct node));

    if(p==NULL)
        return;

    printf("Enter element: ");
    scanf("%d",&p->data);
    p->next=NULL;

    if(rear==NULL)
        front=rear=p;
    else
```

```
    {
        rear->next=p;
        rear=p;
    }
}

void delete()
{
    struct node *t;

    if(front==NULL)
    {
        printf("Queue Underflow");
        return;
    }

    t=front;
    printf("Deleted: %d",t->data);
    front=front->next;

    if(front==NULL)
        rear=NULL;

    free(t);
}

void display()
{
    struct node *t=front;

    while(t!=NULL)
    {
        printf("%d -> ",t->data);
```

```
        t=t->next;
    }

    printf("NULL");
}

void main()
{
    int ch;
    clrscr();

    do
    {
        printf("\n1.Insert\n2.Delete\n3.Display\n4.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1: insert(); break;
            case 2: delete(); break;
            case 3: display(); break;
        }
    }while(ch!=4);

    getch();
}
```

Program 19 — P19_CIRCULAR_QUEUE.C

c

```
#include<stdio.h>
#include<conio.h>
```

```
#define MAX 5

int q[MAX],front=-1,rear=-1;

void insert()
{
    int x;

    if((rear+1)%MAX==front)
    {
        printf("Queue Overflow");
        return;
    }

    printf("Enter element: ");
    scanf("%d",&x);

    if(front==-1)
        front=rear=0;
    else
        rear=(rear+1)%MAX;

    q[rear]=x;
}

void delete()
{
    if(front==-1)
    {
        printf("Queue Underflow");
        return;
    }
```

```
    printf("Deleted: %d",q[front]);

    if(front==rear)
        front=rear=-1;
    else
        front=(front+1)%MAX;
}

void display()
{
    int i;

    if(front==-1)
    {
        printf("Queue empty");
        return;
    }

    i=front;

    while(1)
    {
        printf("%d ",q[i]);

        if(i==rear)
            break;

        i=(i+1)%MAX;
    }
}

void main()
{
```

```
    int ch;
    clrscr();

    do
    {
        printf("\n1.Insert\n2.Delete\n3.Display\n4.Exit\n");
        scanf("%d",&ch);

        switch(ch)
        {
            case 1: insert(); break;
            case 2: delete(); break;
            case 3: display(); break;
        }
    }while(ch!=4);

    getch();
}
```

Program 20 — P20_BST_INORDER.C

c

```
#include<stdio.h>
#include<conio.h>
#include<stdlib.h>

struct node
{
    int data;
    struct node *left,*right;
};
```

```
struct node *create(int x)
{
    struct node *p;
    p=(struct node*)malloc(sizeof(struct node));

    if(p==NULL)
        exit(1);

    p->data=x;
    p->left=NULL;
    p->right=NULL;

    return p;
}

struct node *insert(struct node *root,int x)
{
    if(root==NULL)
        return create(x);

    if(x<root->data)
        root->left=insert(root->left,x);
    else if(x>root->data)
        root->right=insert(root->right,x);

    return root;
}

void inorder(struct node *root)
{
    if(root!=NULL)
    {
        inorder(root->left);
```

```
        printf("%d ",root->data);
        inorder(root->right);
    }
}

void main()
{
    struct node *root=NULL;
    int n,i,x;
    clrscr();

    printf("Enter number of nodes: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
    {
        printf("Enter value: ");
        scanf("%d",&x);
        root=insert(root,x);
    }

    printf("In-order traversal: ");
    inorder(root);

    getch();
}
```

Program 21 — P21_ARRAY_LINEAR_SEARCH.C

c

```
#include<stdio.h>
#include<conio.h>
```

```
void main()
{
    int a[20],n,key,i,found=0;
    clrscr();

    printf("Enter array size: ");
    scanf("%d",&n);

    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter element to search: ");
    scanf("%d",&key);

    for(i=0;i<n;i++)
    {
        if(a[i]==key)
        {
            printf("Found at position %d",i+1);
            found=1;
            break;
        }
    }

    if(found==0)
        printf("Element not found");

    getch();
}
```

Program 22 — P22_ARRAY_INSERT.C

c

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int a[20],n,i,pos,value;
    clrscr();

    printf("Enter number of elements (maximum 19): ");
    scanf("%d",&n);

    if(n<0 || n>=20)
    {
        printf("Invalid size");
        getch();
        return;
    }

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter position: ");
    scanf("%d",&pos);

    if(pos<1 || pos>n+1)
    {
        printf("Invalid position");
        getch();
        return;
    }
```

```
    printf("Enter value: ");
    scanf("%d",&value);

    for(i=n;i>=pos;i--)
        a[i]=a[i-1];

    a[pos-1]=value;
    n++;

    printf("Array after insertion:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    getch();
}
```

Program 23 — P23_ARRAY_DELETE.C

c

```
#include<stdio.h>
#include<conio.h>

void main()
{
    int a[20],n,i,pos;
    clrscr();

    printf("Enter number of elements: ");
    scanf("%d",&n);

    if(n<1 || n>20)
```

```
    {
        printf("Invalid size");
        getch();
        return;
    }

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter position to delete: ");
    scanf("%d",&pos);

    if(pos<1 || pos>n)
    {
        printf("Invalid position");
        getch();
        return;
    }

    for(i=pos-1;i<n-1;i++)
        a[i]=a[i+1];

    n--;

    printf("Array after deletion:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    getch();
}
```

Important: These are traditional Turbo C-style programs. Check array sizes before entering data, enter sorted values for binary search, and compile each file separately.